Shubhadeep Mandal

81602 63797 shubhadeepmandal11@gmail.com shubhadeep-mandal Seronic2001

Education

International Institute of Information Technology, Hyderabad July 2025 – Present
Telangana, India
Master of Technology in Computer Science and Engineering *CGPA: 9.12*

Indian Institute of Information Technology, Ranchi Aug 2020 – Jun 2024
Ranchi, India
Bachelor of Technology (Hons.) in Computer Science and Engineering *CGPA: 9.05*

Projects

Harpy: AI Competitive Programming Toolkit | *Python, MCP Server, C++, Subprocesses, CI/CD*

- Architected an open-standard JSON-RPC 2.0 stdio MCP server exposing 4 deterministic tools, connecting LLM coding agents (Cursor, Antigravity, Claude) to local C++ compiler toolchains and CPH test runners.
- Engineered automated brute-force differential testing and a sandboxed multi-language runner with sub-millisecond execution benchmarking, TLE watchdog interrupts, and side-by-side terminal diffs.
- Implemented dual-channel CPH synchronization via raw HTTP sockets and offline `.cph` test files, distributed as a PyPI package and PyInstaller standalone binary backed by Python 3.10–3.12 CI.

IndicLLM: Transformer Pretraining & Reasoning SFT | *PyTorch, SentencePiece, ONNX Runtime Web*

- Built Hindi and Assamese LMs end to end: curated 1.25B tokens (20%+ self-crawled and OCR'd), trained native 16K BPE tokenizers, and pretrained four 25M-param Transformers from PyTorch primitives on 500M tokens each.
- Fine-tuned on solver-generated reasoning, reaching **0.96** test accuracy and **0.85** on 10-hop chains (3× the training depth) vs a 0.48 baseline, with a leakage check ruling out answer lookup.
- Exported all 10 models to ONNX with a KV cache and a JavaScript tokenizer so they run fully in the browser, deployed as a live demo on Railway via Docker.

BoardBuddy: Real-Time Collaborative Canvas | *Next.js 16, Convex DB, Liveblocks, TypeScript*

- Architected a collaborative workspace supporting real-time multi-user syncing, freehand sketching, and vector shapes.
- Synchronized canvas state, live cursors and in-progress pen strokes across concurrent sessions using Liveblocks and Convex DB.
- Enforced multi-tenant isolation by authorizing every board read and write against the server-verified organization claim, never client-supplied IDs, and gating each live-collaboration room behind an ownership check.

Peer-to-Peer Distributed File Sharing System | *C++, POSIX Sockets, Multithreading, OpenSSL, Linux*

- Built a BitTorrent-style P2P system in C++; self-scheduling workers cut 1GB single-peer downloads from **193s to under 13s**.
- Implemented out-of-order parallel chunk writing directly to designated file offsets and dynamic bitfield generation for real-time partial seeding during active downloads.
- Architected Primary-Backup tracker failover with race-free election and state sync; **46 unit tests** pass under ASan/UBSan.

Technical Skills

Languages: C/C++, Rust, Python, TypeScript, JavaScript, SQL, Bash.

AI Systems & LLMs: Model Context Protocol (MCP), Transformers (Pretrain & SFT), FlashAttention-2, RoPE, PyTorch.

Systems & Tooling: Subprocess Sandboxing, POSIX Sockets, Multithreading, Linux/Unix, Docker, Git, CI/CD.

Full-Stack & Cloud: Next.js, React, Node.js, Convex DB, Prisma ORM, PostgreSQL, WebSockets, REST APIs.

Achievements

- **B.Tech Rank Holder:** Ranked 2nd in IIIT Ranchi by GPA in B.Tech CSE.
- **PGEE 2025:** Secured All-India Rank 100 in IIIT-Hyderabad Post Graduate Entrance Examination (PGEE).
- **Megathon 2025:** Third rank team in Megathon 2025 in the mobile computing track.

Interests

- Web Novel and Manga Enthusiast.
- Chess and Badminton.